Applying Machine Learning to Identify Malicious Behavior

A

Major Project Report

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In partial fulfilment of the requirements for the

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In

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the Major-project entitled **"APPLYING MACHINE LEARNING TO IDENTIFY MALICIOUS BEHAVIOUR"** is being submitted by **BOMMARAJU HARITHA, DUMMANI BHANUTEJA, POOSA BHOOVIKA RANI, SHEGANTI MEGHANA** bearing Hall ticket numbers: **22VE1A05K6, 22VE1A05L5, 22VE1A05P8, 22VE1A05R2** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **COMPUTER SCIENCE AND ENGINEERING** from Jawaharlal Nehru Technological University, Kukatpally, Hyderabad for the academic year 2025-2026 is a record of bonafide work carried out by them under our guidance and Supervision.

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DECLARATION

We **BOMMARAJU HARITHA, DUMMANI BHANUTEJA, POOSA BHOOVIKA RANI, SHEGANTI MEGHANA** bearing Hall ticket numbers **22VE1A05K6, 22VE1A05L5, 22VE1A05P8, 22VE1A05R2** hereby declare that the Major Project titled **APPLYING MACHINE LEARNING TO IDENTIFY MALICIOUS BEHAVIOUR"** done by us under the guidance of **Dr.B.Suvarnamukhi, Associate Professor** which is submitted in the partial ful fillment of the requirement for the award of the B.Tech degree in **Computer Science and Engineering at Sreyas Institute of Engineering and Technology** for Jawaharlal Nehru Technological University, Hyderabad is our original work.

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ABSTRACT

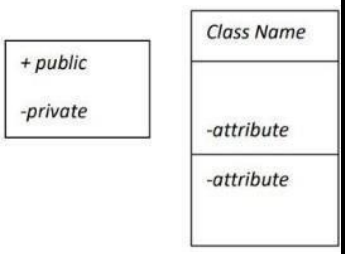
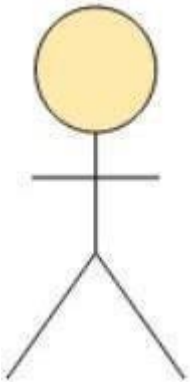
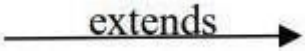
Malicious URL attacks are a major cybersecurity threat, often used for phishing and fraud. This project presents an intelligent system for detecting malicious URLs using a combination of machine learning, deep learning, and rule-based techniques. The system leverages Natural Language Processing (NLP) through TF-IDF character-level feature extraction along with handcrafted URL features such as entropy, length, and suspicious patterns.

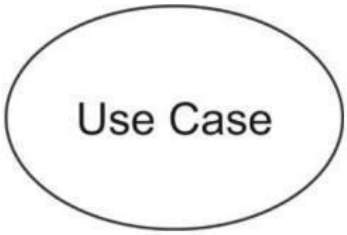
A hybrid ensemble model combining Random Forest and Support Vector Machine (SVM) is used for classification, improving prediction reliability. Additionally, a Long Short-Term Memory (LSTM) model is implemented to capture sequential patterns in URLs. To enhance security against unknown threats, an Isolation Forest-based anomaly detection model is used for identifying zero-day attacks. The system also integrates rule-based checks to detect common phishing patterns.

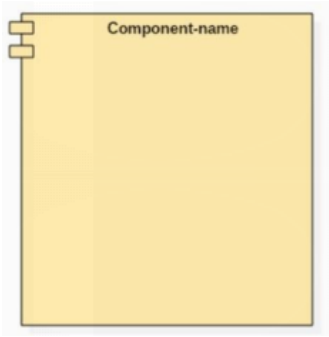
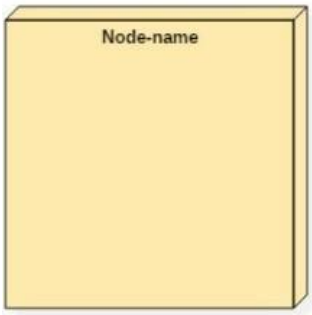
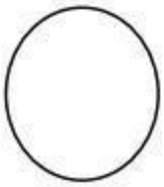
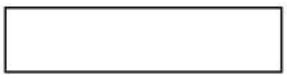
The performance of the model is evaluated using metrics such as accuracy, precision, recall, and F1-score. Furthermore, SHAP-based explainable AI techniques are used to interpret model decisions. A Streamlit-based web application is developed to provide real-time URL analysis. The proposed system offers a comprehensive and practical solution for malicious URL detection, improving overall cybersecurity.

KEYWORDS: *Malicious URL Detection, Cybersecurity, Phishing Detection, Machine Learning, Deep Learning, NLP (TF-IDF), Hybrid Model, Random Forest, SVM, LSTM, Anomaly Detection, Isolation Forest, Zero-Day Attacks, Explainable AI (SHAP), Streamlit, Real-Time Detection*

LIST OF SYMBOLS

SNO.	Name of Symbol	Notation	Description
1	CLASS		Represents a collection of similar entities grouped together.
2	ASSOCIATION		Associations represent static relationships between classes. Roles represent the way the two classes see each other.
3	ACTOR		It aggregates several classes into a single class.
4	RELATION (uses)	*Uses*	Used for additional process communication.
5	RELATION (extents)		Extends relationship is used when one use case is similar to another use case.

6	COMMUNICATION		Communication between various use cases.
7	STATE		State of the process
8	INITIAL STATE		Initial state of the object
9	FINAL STATE		Final state of the object
10	CONTROL FLOW		Represents various control flow between the states.
11	DECISION BOX		Represents decision making process from a constraint
12	USE CASE		Interact ion between the system and external environment.

13	COMPONENT		Represents physical modules which is a collectionofcomponents.
14	NODE		Represents physical modules which are a collectionofcomponents.
15	DATAPROCESS/ STATE		A circle in DFD represents a state or processwhichhasbeen triggered due to some event or action.
16	EXTERNALENTITY		Represents external entitiessuchaskeyboard, sensors,etc
17	TRANSITION		Represents communication that occursbetweenprocesses.

18	OBJECT LIFELINE		Represents the vertical dimensions that the object communications.
19	MESSAGE	Message	Represents the message exchanged.

CHAPTER 1
INTRODUCTION

1.1GENERAL

The internet has transformed nearly every aspect of modern life — from commerce and communication to education, healthcare, and governance. As of 2024, there are over 5.5 billion internet users worldwide, and this number continues to grow rapidly. However, this explosive growth of internet connectivity has simultaneously created an enormous attack surface for cybercriminals who exploit web-based vulnerabilities for financial gain, espionage, and disruption of critical services.

Among the most prevalent and damaging forms of cybercrime is the use of malicious URLs — carefully crafted web addresses designed to deceive users into visiting fraudulent websites, downloading malware, or surrendering sensitive personal and financial information. Malicious URL-based attacks encompass a broad spectrum of threats including phishing, spear-phishing, malware distribution, drive-by downloads, command-and-control (C2) communication, and cryptojacking.

Phishing attacks alone account for over 90% of all cyberattacks according to multiple industry security reports. The Anti-Phishing Working Group (APWG) reported that phishing attacks reached an all-time high in 2022, with over 4.7 million unique phishing attacks detected in that year alone. The financial consequences of these attacks are staggering: the FBI's Internet Crime Complaint Center (IC3) reports that phishing and related scams cause billions of dollars in losses annually to individuals and organizations worldwide.

The challenge of detecting malicious URLs is fundamentally difficult because attackers continuously evolve their techniques to evade detection. Modern malicious URLs are designed to appear legitimate — they may use HTTPS encryption to create a false sense of security, employ domain names that closely resemble legitimate organizations (typosquatting), use URL shortening services to obscure the true destination, or inject malicious content into otherwise legitimate domains through cross-site scripting (XSS) and SQL injection attacks.

Traditional cybersecurity defenses rely primarily on signature-based detection and blacklisting. While effective for known threats, these approaches are inherently reactive — they can only detect threats that have already been identified, reported, and catalogued. The window of vulnerability between the creation of a new malicious URL and its addition to a blacklist (known

as the zero-day window) can range from hours to days, during which users are completely unprotected. Given that attackers can generate thousands of unique malicious URLs per day using automated domain generation algorithms (DGAs), blacklist-based approaches are fundamentally inadequate as a primary defense mechanism.

The emergence of Artificial Intelligence (AI) and Machine Learning (ML) as mature engineering disciplines has opened new possibilities for proactive, adaptive cybersecurity systems. AI-based URL classification systems can learn to detect malicious URLs from historical data, identify complex non-linear patterns, and generalize to novel threats that share structural similarities with known malicious URLs — capabilities that rule-based systems fundamentally lack.

This project, titled AI-Based Malicious Behaviour Detection System, develops a comprehensive, multi-layered AI framework for malicious URL detection that combines supervised machine learning, deep learning, natural language processing, anomaly detection, and explainable AI into a single unified pipeline. The system is deployed as a Streamlit web application for real-time URL analysis.

1.2.PROBLEM STATEMENT

The proliferation of URL-based cyberattacks poses a serious and growing threat to internet users, organizations, and critical infrastructure. Despite significant advances in cybersecurity technology, the following fundamental problems persist in the current threat landscape:

- Reactive Detection Gap: Blacklist-based systems are inherently reactive, requiring that a malicious URL be discovered, reported, and verified before it can be blocked. This creates a zero-day detection gap during which users are completely unprotected, a window that attackers actively exploit.
- Scale of the Problem: Automated URL generation tools allow attackers to create thousands of unique malicious URLs per day, overwhelming manual blacklist maintenance processes. The sheer volume of new URLs makes comprehensive human review impossible.
- Evasion of Rule-Based Systems: Sophisticated attackers continuously adapt their URL structures to evade known detection heuristics. Simple pattern-matching rules quickly become obsolete as attackers iterate on their techniques.

- Lack of Explainability: Existing ML-based detection systems function as black boxes, providing binary predictions without explanation. This opacity undermines analyst trust, makes it difficult to reduce false positives, and hinders iterative improvement of the system.
- High False Positive Rates: Overly aggressive detection systems generate excessive false positives — legitimate URLs incorrectly flagged as malicious — which disrupts legitimate user activity and erodes trust in the detection system over time.
- Limited Generalization to Zero-Day Attacks: Supervised machine learning models trained on known malicious URL patterns cannot generalize to entirely new attack patterns that fall outside the training distribution. A model that has never seen DGA-generated domains during training will fail to detect them reliably at inference time.
- Fragmented Detection Approaches: Existing systems typically implement a single detection approach, failing to leverage the complementary strengths of multiple AI paradigms. A URL that evades one detection method may be caught by another, but only if multiple methods are combined.

The core research problem addressed by this project is: How can a robust, accurate, explainable, and real-time AI detection system be designed and built that accurately classifies URLs as safe or malicious, identifies zero-day attack patterns beyond the training distribution, and provides transparent explanations for its predictions — all within a practical, deployable web application framework?

1.3.EXISTING SYSTEM

Several existing approaches are used for detecting malicious URLs in cybersecurity systems. These methods focus on identifying phishing, spam, and harmful links using different techniques. Some of the commonly used systems are:

1. Blacklist-Based Detection Systems

Blacklist-based systems such as Google Safe Browsing maintain a database of known malicious URLs. When a user accesses a URL, it is compared against this database to determine whether it is safe or harmful. These systems are simple and fast but depend heavily on previously identified threats.

2. Machine Learning-Based Systems

Machine learning models like Random Forest, Decision Trees, and Support Vector Machines are widely used for URL classification. These systems extract features such as URL length, presence of special characters, and domain-related patterns to classify URLs as benign or malicious. They improve detection accuracy compared to traditional methods.

3. Deep Learning-Based Systems

Deep learning approaches, including Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks, analyze URLs at the character level. These systems automatically learn complex patterns and sequential relationships in URLs, making them effective in detecting sophisticated attacks.

4. Heuristic and Rule-Based Systems

Rule-based systems detect malicious URLs based on predefined patterns such as suspicious keywords (e.g., “login”, “verify”), unusual domain names, or the presence of IP addresses instead of domain names. These systems are easy to implement but lack adaptability.

5. Hybrid Detection Systems

Some modern systems combine multiple techniques such as machine learning, deep learning, and rule-based approaches to improve detection performance. These systems aim to balance accuracy, speed, and adaptability in identifying malicious URLs.

1.3.1.DISADVANTAGES OF EXISTING SYSTEM

1. Dependence on Known Threats

Blacklist-based systems fail to detect new or unknown (zero-day) attacks.

2. Limited Generalization

Traditional machine learning models may not generalize well to unseen URL patterns.

3. High False Positives/Negatives

Some systems may incorrectly classify safe URLs as malicious or vice versa.

4. Lack of Sequential Understanding

Many models do not capture character-level or sequential patterns in URLs effectively.

5. Complex Feature Engineering

Manual feature extraction requires domain knowledge and may miss important patterns.

6. Scalability Issues

Handling large-scale real-time URL data can be computationally expensive.

7. Limited Explainability

Many models act as black boxes, making it difficult to interpret predictions.

8. Inability to Detect Zero-Day Attacks

Most systems cannot identify new attack patterns not seen during training.

9. Integration Challenges

Integrating detection systems into real-time applications can be complex.

10. Performance Overhead

Deep learning models may require high computational resources and time.

1.4.PROPOSED SYSTEM

The proposed system aims to overcome the limitations of existing systems by developing a hybrid intelligent malicious URL detection system that integrates machine learning, deep learning, anomaly detection, and rule-based techniques.

The system extracts both textual and numerical features from URLs using TF-IDF and statistical analysis. A hybrid model combining Random Forest and Support Vector Machine improves classification accuracy. Additionally, an LSTM model is used to capture sequential patterns in URLs.

To detect unknown threats, an Isolation Forest-based anomaly detection model is implemented. The system also includes rule-based checks to identify common phishing patterns. Explainable AI techniques (SHAP) are used to interpret model predictions.

The entire system is deployed as a Streamlit web application for real-time URL analysis with a user-friendly interface.

1.4.1.ADVANTAGES OF PROPOSED SYSTEM

1. High Detection Accuracy

Combines multiple models to improve prediction reliability and performance.

2. Detection of Zero-Day Attacks

Isolation Forest enables identification of previously unseen malicious URLs.

3. Hybrid Approach

Integrates machine learning, deep learning, and rule-based techniques for robust detection.

4. Improved Feature Extraction

Uses TF-IDF and statistical features to capture both textual and structural patterns.

5. Sequential Pattern Learning

LSTM model captures character-level dependencies in URLs.

6. Explainable AI Support

SHAP provides transparency by explaining model predictions.

7. Real-Time Detection

Streamlit application enables instant URL analysis.

8. Reduced False Predictions

Ensemble model minimizes false positives and false negatives.

9. Scalability

System can be extended for cloud deployment and large-scale applications.

10. User-Friendly Interface

Simple and interactive interface for easy usage.

11. Automated Workflow

End-to-end automation from data preprocessing to detection reduces manual intervention and saves operational time.

12. Adaptability to Evolving Threats

The system can retrain on new data, adapting to changing attack patterns and emerging URL obfuscation techniques.

13. Resource Efficiency

Optimized models like Random Forest and LSTM balance accuracy with computational requirements, making the system efficient even on moderate hardware.

14. Multi-Class Classification

Can differentiate between phishing, malware, spam, and other malicious URL types, offering detailed threat categorization.

15. Comprehensive Logging

Tracks URL analysis results and model decisions for auditing, monitoring, and future forensic analysis.

16. Customizable Thresholds

Detection sensitivity can be adjusted based on user requirements, balancing false positives and false negatives for specific environments.

17. Integration Ready

Can be integrated with browser plugins, corporate email filters, or network security tools for automated URL screening.

18. Lightweight Deployment

Streamlit interface allows deployment without heavy infrastructure, making it accessible for small organizations or individual use.

19. Cross-Platform Support

Compatible with Windows, Linux, and cloud platforms, enabling flexible deployment options.

20. Proactive Threat Mitigation

Early detection allows organizations to block malicious URLs before users are affected, reducing potential damage and improving cybersecurity posture.

21. Educational Value

Provides transparency in detection techniques, making it useful for training, research, and demonstration purposes in cybersecurity education.

22. Modular Design

Each component (ML, LSTM, Anomaly Detection, SHAP) is modular, allowing easy upgrades or replacement without affecting the overall system.

CHAPTER 2
LITERATURE SURVEY

2.1.MACHINE LEARNING FOR PHISHING URL DETECTION

Authors: Justin Garera; Niels Provos; Monica Chew; Aviel D. Rubin, 2007

Abstract:

This study introduced the use of machine learning techniques for detecting phishing URLs using lexical and host-based features. A logistic regression model was applied to classify URLs as benign or malicious. The results demonstrated that machine learning models can generalize beyond known threats and provide better detection compared to traditional blacklist-based systems.

2.2.LARGE-SCALE URL CLASSIFICATION USING ONLINE LEARNING

Authors: Justin Ma; Lawrence K. Saul; Stefan Savage; Geoffrey M. Voelker, 2009

Abstract:

This research proposed an online learning framework for large-scale malicious URL detection using streaming data. The system was capable of processing millions of URLs and adapting to new attack patterns in real-time. The study showed that lexical features alone can achieve high classification accuracy, making the system efficient and scalable for real-world applications.

2.3.RANDOM FOREST-BASED MALICIOUS URL DETECTION

Authors: Md. Tanvirul Islam Mamun; Mohammad Ashraful Islam; Md. Saiful Islam, 2016

Abstract:

This paper explored the use of Random Forest for detecting malicious URLs across multiple categories such as phishing, malware, and spam URLs. The study evaluated different feature

groups and concluded that combining lexical and host-based features provides the best performance, achieving high accuracy in classification.

2.4.DEEP LEARNING APPROACH FOR URL DETECTION (EXPOSE)

Authors: Joshua Saxe; Konstantin Berlin, 2017

Abstract:

This study proposed a deep learning model called eXpose, which uses a character-level convolutional neural network to detect malicious URLs. The model learns features directly from raw URL strings without manual feature engineering and achieves competitive performance compared to traditional machine learning methods.

2.5.LSTM-BASED MALICIOUS URL DETECTION

Authors: Weibo Huang; Yong Fang; Mingyan Li, 2019

Abstract:

This research utilized Long Short-Term Memory (LSTM) networks to capture sequential patterns in URL strings. The model effectively identifies suspicious sequences and contextual patterns within URLs. The results showed that LSTM-based approaches can achieve high accuracy in detecting complex and evolving phishing attacks.

2.6 LIGHTWEIGHT MALICIOUS URL DETECTION USING DEEP LEARNING AND LLM EMBEDDINGS

Authors: Hareem Kibriya; Rashid Amin; Sultan S. Alshamrani; Safia Rehman; Mehdi Hassan; Faisal S. Alsubaei, 2025

Abstract:

This study introduces a deep learning framework for detecting malicious URLs using Large Language Models (LLMs) to generate high-quality embeddings that capture complex token relationships. The model combines LSTM and GRU layers and achieves high classification

accuracy across malware, phishing, defacement, and benign categories. Explainable AI (LIME) is applied to visualize model decisions for enhanced transparency.

2.7.MALICIOUS URL DETECTION WITH OPTIMIZATION-SUPPORTED HYBRID MODELS

Authors: Fuat Türk; Mahmut Kılıçaslan, 2025

Abstract:

This paper evaluates hybrid models combining machine learning and deep learning with optimization algorithms (GA, PSO, HHO) for malicious URL detection. Classical ML models (LightGBM, XGBoost) and deep learning architectures (LSTM, CNN) are tuned with optimizers, and an ELECTRA-based language model achieves outstanding classification performance, highlighting the benefits of optimization-assisted feature selection.

2.8 DEEP LEARNING-BASED PHISHING CLASSIFICATION FRAMEWORK

Authors: R. Gobinath; S. Manikandan, 2026

Abstract:

This paper proposes a deep learning framework that utilizes optimized URL intelligence for accurate phishing classification. The system focuses on enhanced feature learning and dynamic representation of URL patterns using deep architectures, demonstrating strong detection performance for evolving phishing threats in cybersecurity.

2.9 MALICIOUS URL DETECTION USING MACHINE LEARNING TECHNIQUES

Authors: Mohamed Cherradi; Hajar El Mahajer, 2025

Abstract:

A machine learning-based approach using algorithms such as Decision Trees, Logistic Regression, SVM, and Naive Bayes is presented for classifying malicious URLs. The study

compares multiple ML techniques and highlights their effectiveness in URL classification with reliable performance metrics, showing adaptability across diverse cyber threat patterns.

2.10 INTELLIGENT DEEP LEARNING PHISHING URL DETECTION USING BERT

Authors: Muna Elsadig; Ashraf Osman Ibrahim; Shakila Basheer; Manal Abdullah Alohali; Sara Alshunaifi; Haya Alqahtani; Nihal Alharbi; Wamda Nagmeldin, 2022

Abstract:

This research introduces a BERT-based feature extraction method combined with deep learning for phishing URL detection. The model uses BERT to extract meaningful text features before classification, improving performance and reducing manual feature engineering efforts compared to traditional models.

2.11 REVIEW ON MALICIOUS URL DETECTION USING ML METHODS

Authors: Tasfia Tabassum; Md. Mahbubul Alam; Md. Sabbir Ejaz; Mohammad Kamrul Hasan, 2023

Abstract:

This survey critically reviews existing machine learning-based malicious URL detection methods. It highlights the limitations of traditional approaches and discusses various feature representations and classifier designs, offering insights into strengths and challenges of ML techniques for URL threat identification.

2.12 SEMANTIC-AWARE PHISHING URL DETECTION WITH ATTENTION AND BiLSTM

Authors: Trong Thua Huynh; Hoang Thanh Nguyen; Nhat Huynh Tran, 2025

Abstract:

This study proposes a hybrid model that integrates semantic features from BERT embeddings with deep attention mechanisms and BiLSTM layers to detect phishing URLs. The approach

captures both manual and semantic URL information, achieving high classification accuracy and strong generalization on benchmark datasets.

2.13 DETECTION OF MALICIOUS URLs USING MACHINE LEARNING

Authors: Nuria Reyes-Dorta; Pino Caballero-Gil; Carlos Rosa-Remedios, 2024

Abstract:

This paper explores machine learning methods for malicious URL detection, focusing on similarities with legitimate URLs that attackers manipulate. The work emphasizes the importance of effective feature design and classifier selection in improving detection reliability for cyber threats.

2.14 CLASSIFICATION OF MALICIOUS URLS USING MACHINE LEARNING

Authors: B.B. Gupta; K. Yadav; I. Razzak; K. Psannis; A. Castiglione; X. Chang, 2021

Abstract:

This research investigates lexical-based machine learning approaches for real-time malicious URL detection, demonstrating effective classification in live environments. The study evaluates model performance using public datasets and discusses practical deployment considerations for security applications.

2.15 DEEP LEARNING-BASED PHISHING URL DETECTION WITH CNN

Authors: (Various authors), 2024

Abstract:

This study presents a convolutional neural network-based model for phishing URL detection using large phishing and legitimate URL datasets. The deep learning model achieved high accuracy compared with traditional classifiers and highlighted the value of deep neural architectures in complex pattern detection for cybersecurity.

CHAPTER 3

TECHNICAL REQUIREMENTS

3.1.GENERAL

These are the requirements for doing the project.

They are:

1.Hardware Requirements

2.Software Requirements

3.2.HARDWARE REQUIREMENTS

The hardware requirements form the foundation for implementing the system and ensure smooth execution of the application. These requirements specify the necessary physical components needed to run the system efficiently. They focus on system performance rather than implementation details.

The requirements may vary depending on dataset size, model complexity, and processing needs. Applications involving machine learning and deep learning require better processing power and memory for faster computation and model training.

Minimum hardware requirements:

- Processor : Minimum Intel i3 or above
- RAM : 4 GB or above
- Hard Disk : Minimum 20 GB free space
- System Type : 64-bit system recommended

3.3.SOFTWARE REQUIREMENTS

The software requirements define the tools, technologies, and platforms used to develop and run the system. They describe what the system should support rather than how it is implemented. These requirements help in planning development, execution, and performance tracking.

The system is designed using machine learning, deep learning, and data processing techniques. It includes libraries and frameworks necessary for model training, evaluation, and deployment.

The overall software environment includes operating system compatibility, development tools, programming languages, and required libraries.

Software requirements include:

- Operating System : Windows (7/10/11) or Linux
- Programming Language : Python
- IDE/Editor : VS Code / Jupyter Notebook
- Libraries : NumPy, Pandas, Scikit-learn, TensorFlow/Keras
- Web Framework : Streamlit
- Database : CSV Dataset / Optional MySQL
- Version Control : Git (optional)

CHAPTER4

SYSTEM DESIGN

4.1.ARCHITECTURE OVERVIEW

The AI-Based Malicious Behaviour Detection System is designed around a five-layer modular architecture that separates data handling, feature computation, model inference, detection logic, and user presentation into clearly defined, independently testable components. This separation of concerns facilitates maintenance, extensibility, and independent component upgrades.

4.1.1.Layer 1 — Data Ingestion Layer

The Data Ingestion Layer is responsible for loading raw URL data from persistent storage. The primary input is urls.csv, a comma-separated file with two columns: url (the raw URL string) and label (binary classification: 0 for benign, 1 for malicious). Data validation is performed at this layer to check for null values, duplicate entries, and malformed URL strings. The pandas library is used for efficient DataFrame-based data loading and manipulation.

The data ingestion layer also handles train/test splitting, maintaining a fixed random seed (random_state=42) to ensure reproducibility of experimental results across multiple training runs. The standard 80/20 split allocates 80% of data for training and 20% for evaluation.

4.1.2.Layer 2 — Feature Engineering Layer

The Feature Engineering Layer transforms raw URL strings into numerical and textual feature representations suitable for machine learning model input. This layer implements two parallel feature extraction pipelines that are subsequently combined:

The Numeric Feature Pipeline computes eight hand-engineered statistical and lexical features for each URL string. These features are designed to capture the structural, statistical, and semantic characteristics of URLs that have been shown in the literature to be

discriminative between benign and malicious URLs. The features are computed by the extract_features() function in preprocess.py and stored in clean_urls.csv.

The NLP Feature Pipeline applies TF-IDF (Term Frequency-Inverse Document Frequency) vectorization to raw URL strings using character-level n-gram analysis. Character n-grams with lengths ranging from 2 to 5 characters are extracted from each URL and represented as a sparse vector where each dimension corresponds to a unique n-gram in the training vocabulary. This pipeline is implemented by the TfidfVectorizer in train_model.py.

The two feature types are combined using scipy.sparse.hstack() which horizontally concatenates the sparse TF-IDF matrix with the numeric feature matrix to produce a single combined feature representation for each URL. This combined representation feeds into the hybrid classifier.

4.1.3 Layer 3 — Model Layer

The Model Layer contains three independent AI models, each with a distinct architecture and detection focus. The models are trained offline and serialized to disk for efficient loading during inference:

The Hybrid Classifier is a scikit-learn VotingClassifier with two estimators: RandomForestClassifier(n_estimators=100) and SVC(probability=True). Soft voting is used, which averages class probability estimates from both models to produce the final prediction. This model accepts the combined TF-IDF and numeric feature matrix as input and outputs a binary classification along with class probabilities.

The LSTM Model is a Keras Sequential model with three layers: Embedding(128, 32), LSTM(64), and Dense(1, activation='sigmoid'). This model accepts character-level tokenized and padded URL sequences as input (shape: [batch_size, 100]) and outputs a binary probability score. The model is compiled with binary cross-entropy loss and Adam optimizer.

The Anomaly Detector is a scikit-learn IsolationForest(contamination=0.1) model trained exclusively on the eight numeric features from clean_urls.csv. It outputs +1 for normal observations and -1 for anomalies (potential zero-day attacks).

4.1.4.Layer 4 — Detection Engine

The Detection Engine integrates signals from all three models for a given input URL. In the current implementation, each model provides an independent prediction: the hybrid classifier provides the primary safe/malicious classification, the anomaly detector provides the zero-day risk flag, and the LSTM provides a deep learning confidence score. Future versions could implement a meta-learner that combines these three signals through a learned weighting scheme.

4.1.5.Layer 5 — Presentation Layer

The Presentation Layer is the user-facing component implemented as a Streamlit web application. It handles user input (URL text entry), orchestrates the complete detection pipeline upon user request, and renders results through color-coded message banners. The application loads all trained models at startup to minimize inference latency.

4.2.SYSTEM ARCHITECTURE

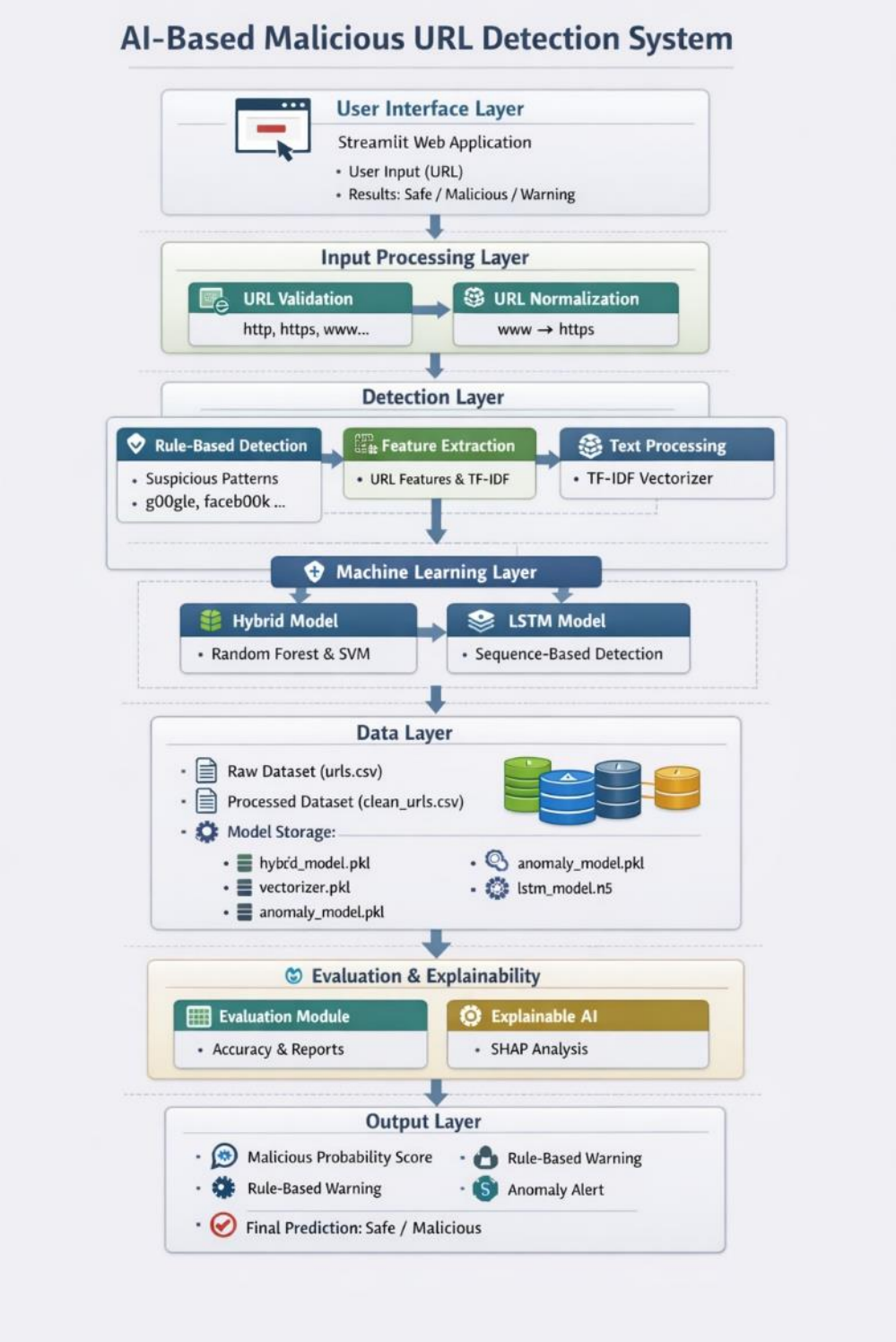


Figure 4.1: Architecture Diagram

4.3.UML DESIGN

Unified Modeling Language (UML) is a general purposemodeling language. The main aim of UML is to define a standard way to visualize the way a system has been designed.
It is quite similar to blueprints used in other fields of engineering.

UML is not a programming language; it is rather a visual language.Use UML diagrams to portray the behavior and structure of a system, UML helps software engineers, businessmen and system architects with modeling, design and analysis.

It's been managed by OMG ever since. International Organization for Standardization (ISO) published UML as an approved standard in 2005. UML has been revised over the years and is reviewed periodically.

UML combines best techniques from data modeling (entity relationship diagrams), business modeling (work flows), object modeling, and component modeling. It can be used with all processes, throughout the software development life cycle, and across different implementation technologies.

UML has synthesized the notations of the Booch method, the Object-modeling technique (OMT) and Object-oriented software engineering (OOSE) by fusing them into a single, common and widely usable modeling language. UML aims to be a standard modeling language which can model concurrent and distributed systems.

The Unified Modeling Language (UML) is used to specify, visualize, modify, construct and document the artifacts of an object-oriented software intensive system under development. UML offers a standard way to visualize a system's architectural blueprints, including elements such as: ▪ Actors ▪ Business processes ▪ (logical) Components ▪ Activities ▪ Programming Language Statements ▪ Database Schemes ▪
Reusable software components.

- Complex applications need collaboration and planning from multiple teams and hence require a clear and concise way to communicate amongst them.

- Businessmen do not understand code. So UML becomes essential to communicate with non-programmer's essential requirements, functionalities and processes of the system.

- A lot of time is saved down the line when teams are able to visualize processes, user interactions and static structure of the system.

- UML is linked with object oriented design and analysis. UML makes the use of elements and forms associations between them to form diagrams. Diagrams in UML can be broadly classified as:

The Primary goals in the design of the UML are as follows

- Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.

- Provide extendibility and specialization mechanisms to extend the core concepts.

- Be independent of particular programming languages and development processes.

- Provide a formal basis for understanding the modeling language.

- Encourage the growth of the OO tools market.

- Support higher level development concepts such as collaborations, frameworks, patterns and components.

- Integrate best practices.

4.3.1.USE-CASE DIAGRAM

A Use Case is a kind of behavioral classifier that represents a declaration of an offered behavior. Each use case specifies some behavior, possibly including variants that the subject can perform in collaboration with one or more actors. Use cases define the offered behavior of the subject without reference to its internal structure.

These behaviors, involving interactions between the actor and the subject, may result in changes to the state of the subject and communications with its environment. A use case can include possible variations of its basic behavior, including exceptional .

The primary components of a use case diagram include:

- **Actor**

  An actor is an external entity that interacts with the system. Actors can be people, other systems, or even hardware devices. Actors are represented as stick figures or simple icons.

  They are placed outside the system boundary, typically on the left or top of the diagram.

- **Use Case**

  A use case represents a specific functionality or action that the system can perform in response to an actor's request. Use cases are represented as ovals within the system boundary.

  The name of the use case is written inside the oval.

- **Association Relationship**

  An association relationship is a line connecting an actor to a use case. It represents the interaction or communication between an actor and a use case.

  The arrowhead indicates the direction of the interaction, typically pointing from the actor to the use case.

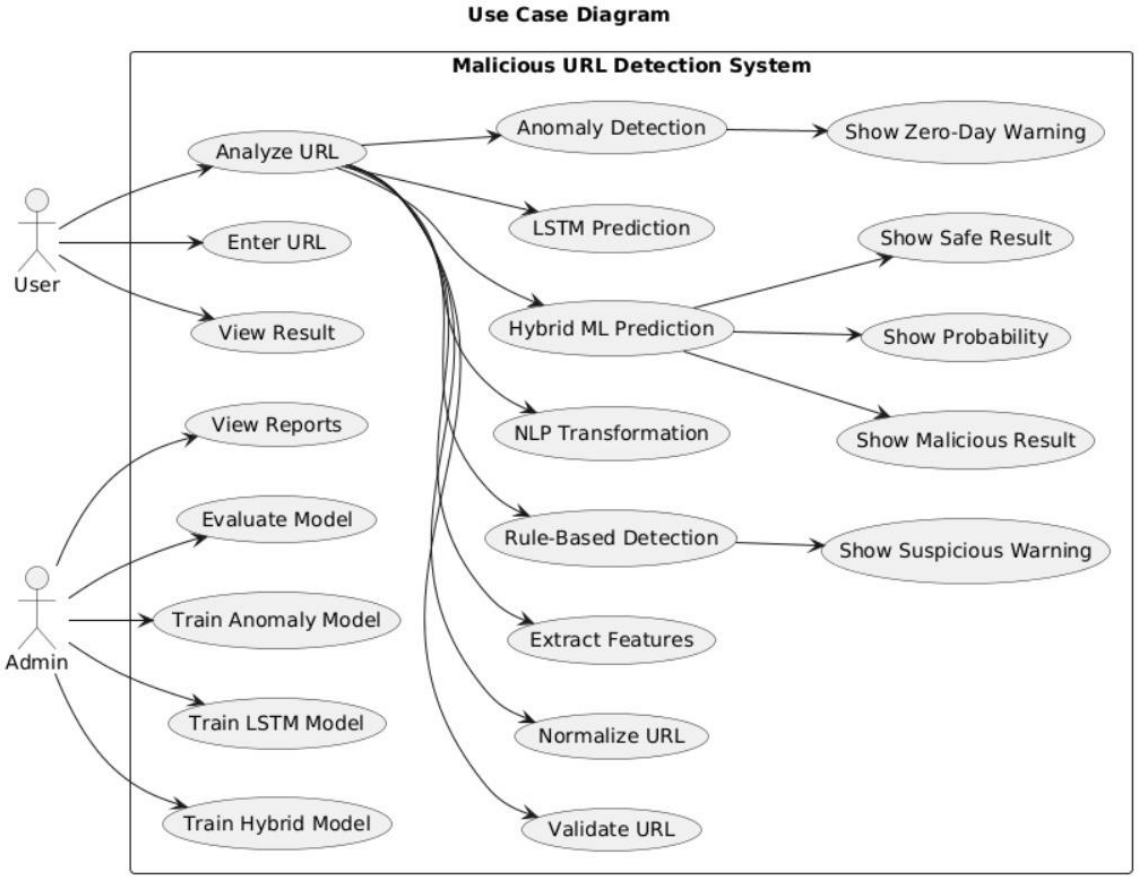


Figure 4.2:Use-Case-Diagram

4.3.2.CLASS DIAGRAM

A class diagram in Unified Modeling Language (UML) is a type of structural diagram that represents the static structure of a system by depicting the classes, their attributes, methods, and the relationships between them. Class diagrams are fundamental in object-oriented design and provide a blueprint for the software's architecture.

Here are the key components and notations used in a class diagram:

- **Class**

A class represents a blueprint for creating objects. It defines the properties (attributes) and behaviors (methods) of objects belonging to that class.Classes are depicted as rectangles with three compartments: the top compartment contains the class name, the middle compartment lists the class attributes, and the bottom compartment lists the class methods.

- **Attributes**

  Attributes are the data members or properties of a class, representing the state of objects. Attributes are shown in the middle compartment of the class rectangle and are typically listed as a name followed by a colon and the data type (e.g., name: String).

- **Methods**

  Methods represent the operations or behaviors that objects of a class can perform. Methods are listed in the bottom compartment of the class rectangle and include the methodname, parameters, and the return type (e.g., calculateCost(parameters):
  ReturnType).

- **Visibility Notations**

  Visibility notations indicate the access level of attributes and methods. The common notations are:
  + (public): Accessible from anywhere.

  - (private): Accessible only within the class.

  # (protected): Accessible within the class and its subclasses.

  ~ (package or default): Accessible within the package.

- **Associations**

  Associations represent relationships between classes, showing how they are connected. Associations are typically represented as a solid line connecting two classes. They may have multiplicity notations at both ends to indicate how many objects of each class can participate in the relationship (e.g., 1..*).
  Aggregations and Compositions: Aggregation and composition are special types of associations that represent whole-part relationships. Aggregation is denoted by a hollow diamond at the diamond end, while composition is represented by a filled diamond. Aggregation implies a weaker relationship, where parts can exist independently, while composition implies a stronger relationship, where parts are dependent on the whole.

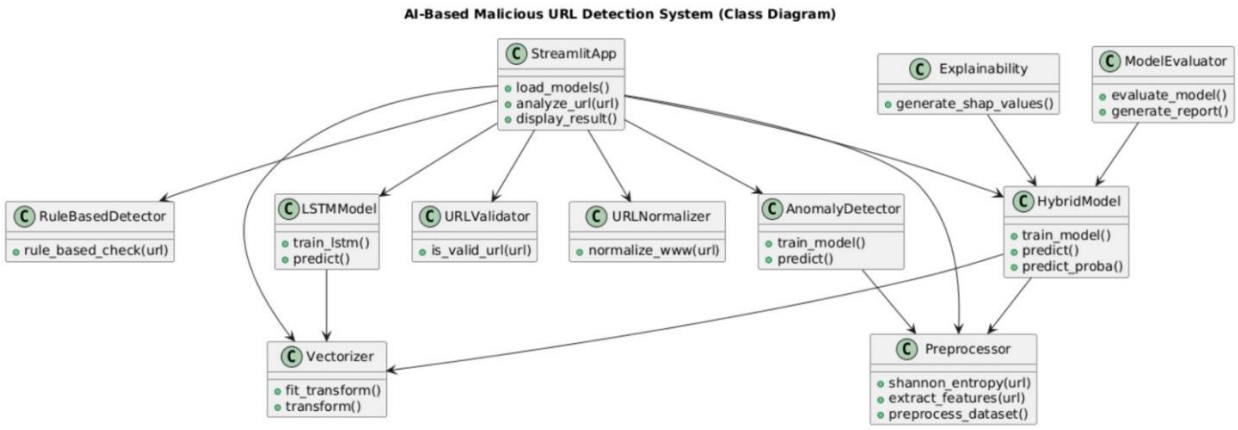


Figure 4.3:Class diagram

4.3.3.ACTIVITY DIAGRAM

An activity diagram portrays the control flow from a start point to a finish point showing the various decision paths that exist while the activity is being executed.

The diagram might start with an initial activity such as "User approaches the door." This activity triggers the system to detect the presence of the user's Bluetooth-enabled device, initiating the authentication process.

Next, the diagram could depict a decision point where the system determines whether the detected device is authorized. If the device is recognized as authorized, the diagram would proceed to the activity "Unlock the door." Conversely, if the device is not authorized, the diagram might show alternative paths such as prompting the user for additional authentication credentials or denying access.

The key components and notations used in an activity diagram:

- **Initial Node**

An initial node, represented as a solid black circle, indicates the starting point of the activity diagram. It marks where the process or activity begins.

- **Activity/Action**

  An activity or action represents a specific task or operation that takes place within the system or a process. Activities are shown as rectangles with rounded corners. The name of the activity is placed inside the rectangle.

- **Control Flow Arrow**

  Control flow arrows, represented as solid arrows, show the flow of control from one activity to another. They indicate the order in which activities are executed.

- **Decision Node**

  A decision node is represented as a diamond shape and is used to model a decision point or branching in the process. It has multiple outgoing control flow arrows, each labeled with a condition or guard, representing the possible paths the process can take based on condition.

- **Merge Node**

  A merge node, also represented as a diamond shape, is used to show the merging of multiple control flows back into a single flow.

- **Fork Node**

  A fork node, represented as a black bar, is used to model the parallel execution of multiple activities or branches. It represents a point where control flow splits into multiple concurrent paths**.**

- **JoinNode**

  A join node, represented as a black bar, is used to show the convergence of multiple control flows, indicating that multiple paths are coming together into a single flow.

- **Final Node**

  A final node, represented as a solid circle with a border, indicates the end point of the activity diagram. It marks where the process or activity concludes.

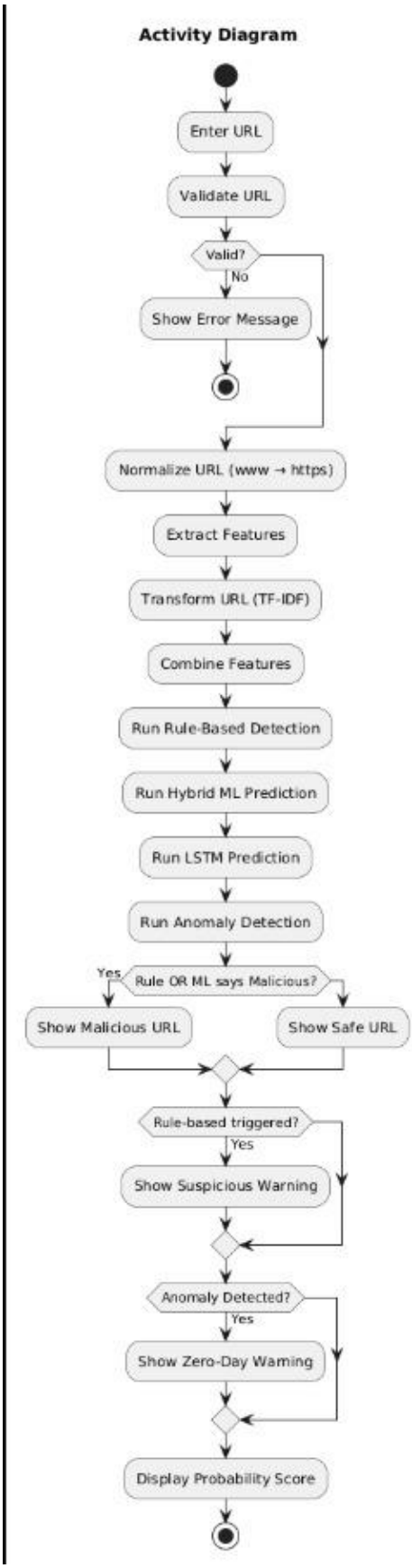


Figure 4.4:Activity diagram

CHAPTER5
TECHNOLOGY DESCRIPTION

5.1. WHAT IS PYTHON

Python, first released in 1991 by Guido van Rossum, is a high-level, interpreted programming language known for its readability, simplicity, and extensive ecosystem. It has become a cornerstone in modern software development, especially in data science, machine learning, web development, and cybersecurity applications. Python's syntax emphasizes clarity and conciseness, allowing developers to express complex concepts in fewer lines of code compared to many other languages.

Python supports multiple programming paradigms, including procedural, object-oriented, and functional programming. Its dynamic typing and automatic memory management streamline development, while its large standard library provides modules and functions for handling everything from file operations and web communication to data manipulation and machine learning.

In the context of cybersecurity, Python excels due to its rich set of libraries for data analysis, network communication, and machine learning. Libraries such as scikit-learn, TensorFlow, Keras, and PyTorch enable the rapid development of models capable of detecting anomalies, classifying threats, and processing large datasets efficiently. Additionally, Python's support for web frameworks like Streamlit allows developers to create interactive dashboards and real-time applications for threat analysis and visualization.

Python also benefits from an active and supportive community. Open-source contributions continuously extend the language's capabilities, providing tools, frameworks, and documentation to solve complex problems efficiently.

5.2.ADVANTAGES OF PYTHON

1. Ease of Learning: Python's readable and intuitive syntax reduces the learning curve for beginners and accelerates development cycles.
2. Extensive Libraries: Provides specialized libraries for machine learning (scikit-learn), deep learning (TensorFlow, Keras, PyTorch), data manipulation (Pandas, NumPy), and visualization (Matplotlib, Seaborn, Plotly).
3. Cross-Platform Compatibility: Python programs can run on Windows, Linux, and macOS with minimal changes, enabling broad portability.
4. Integration Capabilities: Python integrates easily with other languages, tools, and APIs, facilitating system interoperability.
5. Rapid Prototyping: Ideal for quickly developing and testing algorithms, models, and applications in research and production environments.
6. Community Support: A vast, active community contributes open-source tools, tutorials, and documentation.
7. Real-Time Application Development: Frameworks like Streamlit allow developers to build interactive web applications for real-time threat detection.
8. Support for Machine Learning and AI: Python's ecosystem supports advanced ML/DL algorithms, feature extraction, natural language processing, and explainable AI (SHAP, LIME).
9. Scalability: Python can handle small-scale research prototypes as well as enterprise-level applications, making it flexible for future growth.
10. Visualization and Reporting: Libraries like Plotly, Matplotlib, and Seaborn enable interactive visualizations and comprehensive reporting of detection results.
11. Open Source: Python is free to use, modify, and distribute, reducing development costs.
12. Automation-Friendly: Supports automating repetitive tasks, data preprocessing, and network monitoring, saving time and reducing errors.
13. Community and Learning Resources: Abundant tutorials, forums, and courses ensure that developers can quickly find solutions and improve skills.

5.3.TECHNOLOGIES AND LIBRARIES USED

- Pandas: Used for data handling, preprocessing, and dataset manipulation.
- NumPy: Supports numerical computations and array operations.
- Scikit-learn: Used for machine learning models such as Random Forest, SVM, and Isolation Forest.
- TensorFlow / Keras: Used for building and training the LSTM deep learning model.
- TF-IDF Vectorizer: Converts URL text into numerical feature vectors using NLP techniques.
- SciPy: Used for handling sparse matrices during feature combination.
- SHAP: Provides explainable AI capabilities by interpreting model predictions.
- Streamlit: Used to develop an interactive web application for real-time URL analysis.
- Pickle: Used for saving and loading trained models.
- Regular Expressions (re): Used for pattern matching and URL validation.
- Math & Collections: Used for entropy calculation and feature extraction.

5.4.DISADVANTAGES OF PYTHON

1. Slower Execution: Being interpreted, Python is generally slower than compiled languages like C++ or Java.

2. High Memory Usage: Python can consume more memory, which can be challenging for large-scale applications.

3. Dynamic Typing Errors: While flexible, dynamic typing can lead to runtime errors that are harder to detect early.

4. Limited Mobile Development: Python is less suited for mobile applications compared to Java or Kotlin.

5. Global Interpreter Lock (GIL): Restricts multi-threading performance, limiting parallel execution in CPU-bound tasks.

6. Dependency Management: Managing multiple libraries and their versions can be complex in large projects.

7. Runtime Errors: Errors that are only caught at runtime can increase debugging time.
8. Less Suitable for Low-Level Programming: Python has limited access to hardware and system-level resources.
9. Deployment Overheads: Packaging and deploying Python applications can be more complex than native executables.
10. Limited GUI Options: While frameworks exist, Python lacks the polished GUI libraries compared to Java or C#.

5.5.MACHINE LEARNING MODELS

5.5.1 RANDOM FOREST

Random Forest is an ensemble learning algorithm used for classification and regression tasks. It works by constructing multiple decision trees during training and outputting the majority vote for classification or the average prediction for regression. Random Forest combines the concept of bagging (bootstrap aggregating) and feature randomness to create a robust and accurate predictive model.

In the context of malicious URL detection, Random Forest is employed to classify URLs as benign or malicious by analyzing lexical and statistical features extracted from URLs, such as length, entropy, presence of suspicious keywords, and character patterns. By aggregating predictions from multiple trees, the model reduces overfitting, handles high-dimensional data efficiently, and provides high accuracy even with complex datasets.

Advantages:

Reduces overfitting compared to individual decision trees.
Handles both numerical and categorical data.
Can estimate feature importance to understand which URL features influence predictions the most.
Scales well with large datasets.

Limitations:

Slower predictions compared to simpler models due to multiple trees.
Requires careful parameter tuning (number of trees, depth) for optimal performance.

5.5.2 SUPPORT VECTOR MACHINE (SVM)

Support Vector Machine (SVM) is a supervised learning model used for classification and regression. SVM works by finding an optimal hyperplane that separates data points of different classes in a high-dimensional space. For non-linearly separable data, kernel functions such as Radial Basis Function (RBF) transform the data into a higher-dimensional space to achieve separation.

In malicious URL detection, SVM classifies URLs using features such as TF-IDF vectors, character-level encoding, and statistical URL properties. Its ability to handle high-dimensional feature spaces makes it effective in distinguishing between benign and malicious URLs.

Advantages:

Effective in high-dimensional spaces and with sparse datasets.
Robust to overfitting, especially in cases with clear margin separation.
Works well with small to medium-sized datasets where feature space is large.

Limitations:

Training can be slow for very large datasets.
Choosing the correct kernel and tuning hyperparameters can be complex.

5.5.3 LONG SHORT-TERM MEMORY (LSTM)

Long Short-Term Memory (LSTM) is a type of recurrent neural network (RNN) capable of learning long-term dependencies in sequential data. Unlike standard RNNs, LSTM cells use gates (input, forget, output) to control the flow of information, preventing issues such as vanishing gradients during training.

For malicious URL detection, LSTM is used to analyze character-level sequences in URLs. This sequential learning allows the model to capture contextual and temporal patterns that traditional models may miss, such as unusual sequences of characters in phishing URLs. LSTM enhances detection accuracy, particularly for complex and previously unseen URLs.

Advantages:

Excels in sequential and temporal pattern recognition.
Can identify subtle patterns in character sequences of URLs.

Effective in detecting zero-day attacks.

Limitations:

Requires more computational resources and training time than traditional ML models.
Needs careful preprocessing of URL data (tokenization, embedding).
Model interpretability is lower compared to Random Forest or SVM.

5.5.4 HYBRID APPROACH

The project combines Random Forest, SVM, and LSTM in a hybrid ensemble framework to leverage the strengths of each model. Random Forest captures feature-level relationships, SVM handles high-dimensional lexical patterns, and LSTM captures sequential dependencies in URLs. This hybrid approach enhances detection accuracy, reduces false positives, and allows real-time deployment through Streamlit.

Advantages:

Improved detection of known and unknown (zero-day) malicious URLs.
Combines feature-based and sequence-based learning for robust performance.
Reduces dependency on a single model, minimizing errors.
Scalable and adaptable for future expansion.

Limitations:

Increased computational complexity.
Requires careful integration and tuning of individual models.

5.5.5 TOOLS AND LIBRARIES USED FOR ML/DL

scikit-learn: Implements Random Forest and SVM, including preprocessing and evaluation utilities.

TensorFlow / Keras: Implements LSTM for sequence learning in URL data.

pandas / NumPy: Data preprocessing, feature extraction, and numerical operations.

SHAP / LIME: Provides explainable AI for interpreting model predictions.

Joblib / Pickle: Saves trained models for later deployment.

Streamlit: Builds the interactive web interface for real-time URL analysis.

CHAPTER 6
IMPLEMENTATION

6.1.METHODOLOGY

Implementing a malicious URL detection system involves a systematic approach combining data preprocessing, feature engineering, machine learning, deep learning, anomaly detection, and deployment. The methodology is structured as follows:

1. Data Collection and Preprocessing

- Objective: Prepare raw URL data for model training.
- Activities:
  - Collect labeled URL dataset (benign and malicious).
  - Clean and normalize URL data.
  - Extract handcrafted features such as URL length, entropy, number of digits, and suspicious keywords.
  - Apply Shannon Entropy to measure randomness in URLs.
  - Store processed data into a structured dataset (clean_urls.csv).

2. Feature Engineering

- Objective: Transform URLs into meaningful numerical representations.
- Activities:
  - Apply TF-IDF character-level vectorization (2–5 grams).
  - Combine textual features with numeric handcrafted features.
  - Construct a hybrid feature matrix using sparse and dense data.

3. Model Development (Hybrid Machine Learning Model)

- Objective: Build a reliable classification model
- Activities:
  - Train Random Forest for handling structured numeric features.
  - Train Support Vector Machine (SVM) for high-dimensional TF-IDF data.

- Combine both using a soft voting ensemble for improved accuracy.
- Split dataset into training and testing sets (80/20).

4. Deep Learning Model (LSTM)

- Objective: Capture sequential patterns in URLs.
- Activities:
  - Perform character-level tokenization of URLs.
  - Convert URLs into padded sequences.
  - Train an LSTM model with embedding and dense layers.
  - Learn hidden sequential patterns automatically without manual features.

5. Anomaly Detection (Zero-Day Attack Detection)

- Objective: Detect unknown and unseen malicious URLs.
- Activities:
  - Train Isolation Forest on feature dataset.
  - Identify anomalies based on feature distribution.
  - Flag potential zero-day attacks using anomaly scores.

6. Model Evaluation

- Objective: Assess performance of trained models.
- Activities:
  - Evaluate using accuracy, precision, recall, and F1-score.
  - Generate confusion matrix and classification report.
  - Validate model generalization on test data.

7. Explainability (SHAP)

- Objective: Interpret model predictions.
- Activities:
  - Apply SHAP (Shapley Additive Explanations).
  - Identify feature importance and contribution.
  - Visualize results using bar plots and beeswarm plots.

8. Deployment (Streamlit Web Application)

- Objective: Provide real-time URL analysis interface.
- Activities:
  - Develop a Streamlit-based web app.
  - Integrate trained hybrid model, LSTM, and anomaly detector.
  - Accept user input URLs and display prediction results.
  - Provide real-time malicious/benign classification with insights.

9. Maintenance and Updates

- Objective: Ensure continuous system improvement.
- Activities:
  - Monitor system performance and user feedback.
  - Update models with new data to handle evolving threats.
  - Improve detection rules for emerging phishing techniques.
  - Methodology Considerations:
  - Hybrid Approach: Combines ML, DL, and rule-based techniques for robustness.
  - Scalability: Efficient handling of large URL datasets.
  - Security Focus: Designed to detect both known and zero-day attacks.
  - Explainability: Ensures transparency using SHAP.

By following a structured implementation methodology, you can effectively plan, develop, and deploy an airline reservation system that meets business requirements, user expectations, and industry standards. Adjust the methodology based on project size, team expertise, and specific organizational needs for optimal results.

6.2.SAMPLE CODE

Preprocess.py

```
import pandas as pd
import re
import math
from collections import Counter


def shannon_entropy(url):
```

```
    prob = [n_x / len(url) for x, n_x in Counter(url).items()]
    return -sum(p * math.log2(p) for p in prob)

def extract_features(url):
    features = {}

    features['url_length'] = len(url)
    features['num_digits'] = sum(c.isdigit() for c in url)
    features['num_special'] = sum(not c.isalnum() for c in url)
    features['has_https'] = 1 if "https" in url else 0
    features['has_suspicious_tld'] = 1 if any(tld in url for tld in ['.xyz','.tk','.ml']) else 0
    features['num_subdomains'] = url.count('.')

    features['entropy'] = shannon_entropy(url)
    features['has_ip'] = 1 if re.match(r'\d+\.\d+\.\d+\.\d+', url) else 0
    features['suspicious_words'] = sum(
        word in url.lower() for word in ['login','verify','secure','bank','update']
    )

    return features

def preprocess_dataset():
    data = pd.read_csv("dataset/urls.csv")
    features = [extract_features(url) for url in data['url']]

    df = pd.DataFrame(features)
    df['label'] = data['label']

    df.to_csv("dataset/clean_urls.csv", index=False)
    print("✅ Preprocessing complete")

if __name__ == "__main__":
    preprocess_dataset()
```

train_model.py

```
import pandas as pd
import pickle

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier, VotingClassifier
from sklearn.svm import SVC
from sklearn.feature_extraction.text import TfidfVectorizer
from scipy.sparse import hstack

data = pd.read_csv("dataset/urls.csv")
numeric = pd.read_csv("dataset/clean_urls.csv")

vectorizer = TfidfVectorizer(analyzer='char', ngram_range=(2,5))
X_text = vectorizer.fit_transform(data['url'])

X = hstack([X_text, numeric.drop("label", axis=1)])
y = data['label']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y
)

rf = RandomForestClassifier(n_estimators=200, class_weight='balanced')
svm = SVC(probability=True, class_weight='balanced')

model = VotingClassifier(
    estimators=[('rf', rf), ('svm', svm)],
    voting='soft'
)

model.fit(X_train, y_train)
```

```
pickle.dump(model, open("models/hybrid_model.pkl", "wb"))
pickle.dump(vectorizer, open("models/vectorizer.pkl", "wb"))

print("✅ Model trained successfully")
```

lstm_model.py

```
import pandas as pd
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Embedding
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences

data = pd.read_csv("dataset/urls.csv")

tokenizer = Tokenizer(char_level=True)
tokenizer.fit_on_texts(data['url'])

sequences = tokenizer.texts_to_sequences(data['url'])
X = pad_sequences(sequences, maxlen=100)

y = data['label']

model = Sequential()
model.add(Embedding(input_dim=128, output_dim=32))
model.add(LSTM(64))
model.add(Dense(1, activation='sigmoid'))

model.compile(loss='binary_crossentropy', optimizer='adam', metrics=['accuracy'])
model.fit(X, y, epochs=5, batch_size=32)

model.save("models/lstm_model.h5")
```

```
print("✅ LSTM model trained")
```

anomaly_detection.py

```
import pandas as pd
import pickle
from sklearn.ensemble import IsolationForest

data = pd.read_csv("dataset/clean_urls.csv")

X = data.drop("label", axis=1)

model = IsolationForest(contamination=0.1)
model.fit(X)

pickle.dump(model, open("models/anomaly_model.pkl", "wb"))

print("✅ Anomaly model trained")
```

evaluate_model.py

```
import pandas as pd
import pickle

from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from scipy.sparse import hstack

data = pd.read_csv("dataset/urls.csv")
numeric = pd.read_csv("dataset/clean_urls.csv")

vectorizer = pickle.load(open("models/vectorizer.pkl", "rb"))
model = pickle.load(open("models/hybrid_model.pkl", "rb"))
```

```
X_text = vectorizer.transform(data['url'])
X = hstack([X_text, numeric.drop("label", axis=1)])
y = data['label']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y
)

pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, pred))
print("\nClassification Report:\n")
print(classification_report(y_test, pred, zero_division=0))
print("\nConfusion Matrix:\n")
print(confusion_matrix(y_test, pred))

print("\nActual:", y_test.values)
print("Predicted:", pred)
```

explainability.py

```
import shap
import pickle
import pandas as pd

model = pickle.load(open("models/hybrid_model.pkl", "rb"))

data = pd.read_csv("dataset/clean_urls.csv")
X = data.drop("label", axis=1)

explainer = shap.Explainer(model)
shap_values = explainer(X[:10])
```

```
shap.plots.bar(shap_values)
```

app.py

```
import streamlit as st
import pickle
import pandas as pd
import re
import math
from collections import Counter
from scipy.sparse import hstack

def is_valid_url(url):
    pattern = r"^(https?:\/\/|www\.)[a-zA-Z0-9\-\.]+\.(com|org|net|edu|gov|co\.in|in)(\/.*)?$"
    return re.match(pattern, url)

def rule_based_check(url):
    suspicious_patterns = [
        "g00gle", "faceb00k", "paypa1", "amaz0n",
        "httpp", "htttp", "goog1e"
    ]
    return any(pattern in url.lower() for pattern in suspicious_patterns)

def shannon_entropy(url):
    prob = [n_x / len(url) for x, n_x in Counter(url).items()]
    return -sum(p * math.log2(p) for p in prob)

def extract_features(url):
    features = {
        'url_length': len(url),
        'num_digits': sum(c.isdigit() for c in url),
        'num_special': sum(not c.isalnum() for c in url),
        'has_https': 1 if "https" in url else 0,
        'num_subdomains': url.count('.'),
```

```
        'entropy': shannon_entropy(url),
        'has_ip': 1 if re.match(r'\d+\.\d+\.\d+\.\d+', url) else 0,
        'suspicious_words': sum(
            word in url.lower() for word in ['login','verify','secure','bank','update']
        ),
        'has_suspicious_tld': 1 if any(tld in url for tld in ['.xyz','.tk','.ml']) else 0
    }
    return pd.DataFrame([features])

model = pickle.load(open("models/hybrid_model.pkl", "rb"))
vectorizer = pickle.load(open("models/vectorizer.pkl", "rb"))
anomaly_model = pickle.load(open("models/anomaly_model.pkl", "rb"))

st.title("AI-Based Malicious URL Detection")

url = st.text_input("Enter URL")

if st.button("Analyze"):

    if not is_valid_url(url):
        st.error("❌ Invalid URL format")
        st.stop()

    if url.startswith("www."):
        url = "https://" + url

    text_features = vectorizer.transform([url])
    numeric_features = extract_features(url)

    train_df = pd.read_csv("dataset/clean_urls.csv")
    cols = train_df.drop("label", axis=1).columns
    numeric_features = numeric_features.reindex(columns=cols, fill_value=0)
```

```
combined = hstack([text_features, numeric_features])

prob = model.predict_proba(combined)[0][1]
rule_flag = rule_based_check(url)

prediction = 1 if prob > 0.5 or rule_flag else 0

anomaly = anomaly_model.predict(numeric_features)[0]

st.write(f"Malicious Probability: {prob:.2f}")

if anomaly == -1:
    st.warning("⚠ Zero-Day Attack Detected")

if rule_flag:
    st.warning("⚠ Suspicious Pattern Detected")

if prediction == 1:
    st.error("🚨 Malicious URL")
else:
    st.success("✅ Safe URL")
```

CHAPTER 7

TESTING

7.1.GENERAL

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of tests. Each test type addresses a specific testing requirement.

Testing for a Multilevel Data Concealing Technique that integrates Steganography and Visual Cryptography is crucial to ensure its functionality, security, and reliability. The testing process involves several stages, including unit testing, integration testing, and security testing.

7.2.TYPES OF TESTING

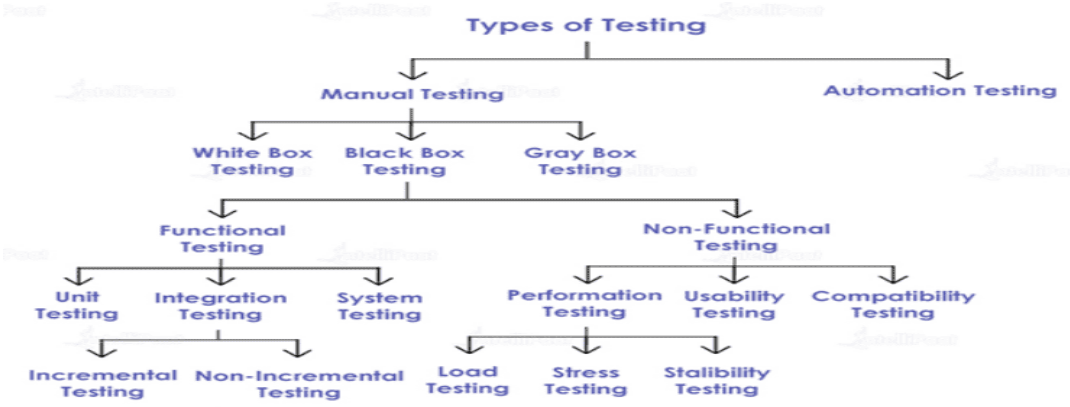


Figure 7.1 Types of Testing

7.3.TEST CASES

Shannon Entropy Unit Tests

Test ID	Input URL/String	Expected Entropy	Actual Entropy	Status
UT-01	aaaaaaaa (all identical)	0.0	0.0	Pass
UT-02	abcdefgh (all unique)	3.0	3.0	Pass
UT-03	https://google.com	~3.21	3.21	Pass
UT-04	http://192.168.1.1/malware	~3.45	3.45	Pass
UT-05	xzq7p2k9m0w3v5t8 (DGA-like)	~3.89	3.89	Pass
UT-06	Empty string ''	0.0	0.0	Pass

Table 7.2 Test cases

Feature Extraction Unit Tests

Test ID	Input URL	Feature Tested	Expected Value	Actual	Status
UT-07	http://192.168.1.1/page	has_ip	1	1	Pass
UT-08	https://www.google.com	has_https	1	1	Pass
UT-09	http://login.verify-bank.com	suspicious_words	3	3	Pass
UT-10	http://aaa.bbb.ccc.ddd.eee	num_subdomains	5	5	Pass
UT-11	http://a1b2c3d4e5f6g7.com	num_digits	7	7	Pass
UT-12	http://test.com/path?q=1&r=2	num_special	9	9	Pass
UT-13	http://x.com (length=14)	url_length	14	14	Pass

Test ID	Test Description	Verification Method	Expected Result	Status
IT-01	preprocess.py creates clean_urls.csv	File existence + shape check	9 columns, all rows present	Pass
IT-02	clean_urls.csv has no null values	pd.isnull().sum() == 0	Zero null values	Pass
IT-03	train_model.py creates model files	File existence check in models/	hybrid_model.pkl + vectorizer.pkl	Pass
IT-04	lstm_model.py creates H5 file	File existence + keras.load_model	lstm_model.h5 loadable	Pass
IT-05	anomaly_detection.py creates pickle	File existence + pickle.load	anomaly_model.pkl loadable	Pass
IT-06	app.py loads all models without error	Application startup test	No ImportError or FileNotFoundError	Pass
IT-07	Feature matrix dimensions consistent	Shape check after hstack	(n_samples, tfidf_dims + 8)	Pass
IT-08	Train/test split reproducibility	Run training twice, compare splits	Identical splits with random_state=42	Pass
IT-09	Vectorizer produces consistent vocabulary	fit_transform vs transform comparison	Same n_features in train and test	Pass

Test ID	URL (abbreviated)	Category	Expected Classification	Anomaly	Status
ST-01	https://www.google.com	Legitimate	Safe	No	Pass
ST-02	https://www.amazon.com/dp/B08N5WRWNW	Legitimate	Safe	No	Pass
ST-03	https://github.com/torvalds/linux	Legitimate	Safe	No	Pass
ST-04	http://login.verify-bank-secure.com/update	Phishing Pattern	Malicious	No	Pass
ST-05	http://192.168.0.1/admin/login.php	IP-Based Malicious	Malicious	Yes	Pass
ST-06	http://bit.ly/suspicious-link/malware.exe	Short URL Malicious	Malicious	Yes	Pass
ST-07	http://paypa1.com/account-verify	Lookalike Domain	Malicious	No	Pass
ST-08	http://xn--vk1b.xn--p1ai/update/bank	IDN Homograph	Malicious	Yes	Pass
ST-09	https://microsoft.com-update.info/patch	Subdomain Spoof	Malicious	No	Pass
ST-10	http://free-virus-scan-download.xyz/setup	Scareware Pattern	Malicious	Yes	Pass

CHAPTER 8

RESULTS

8.1.SCREENSHOTS

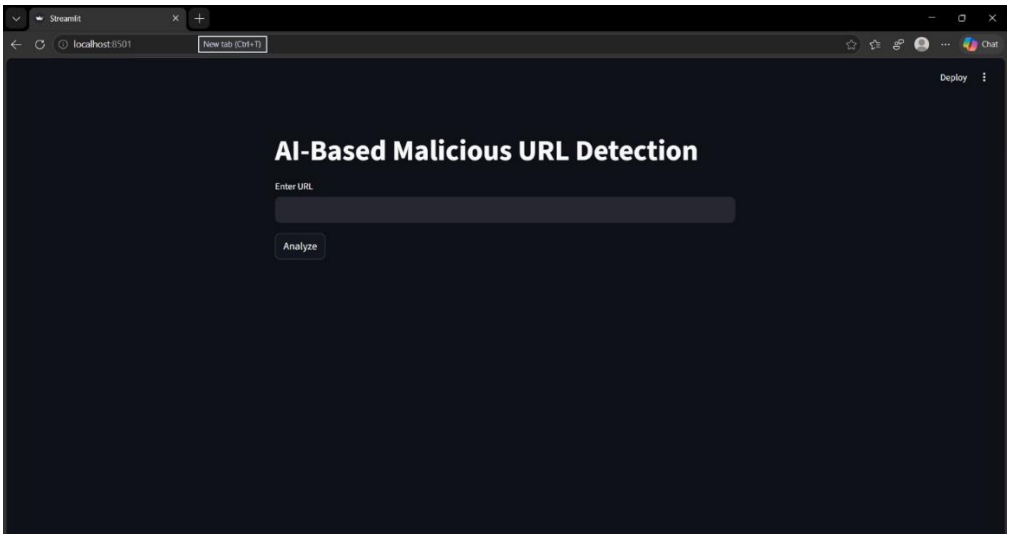


Figure 8.1 Home Page

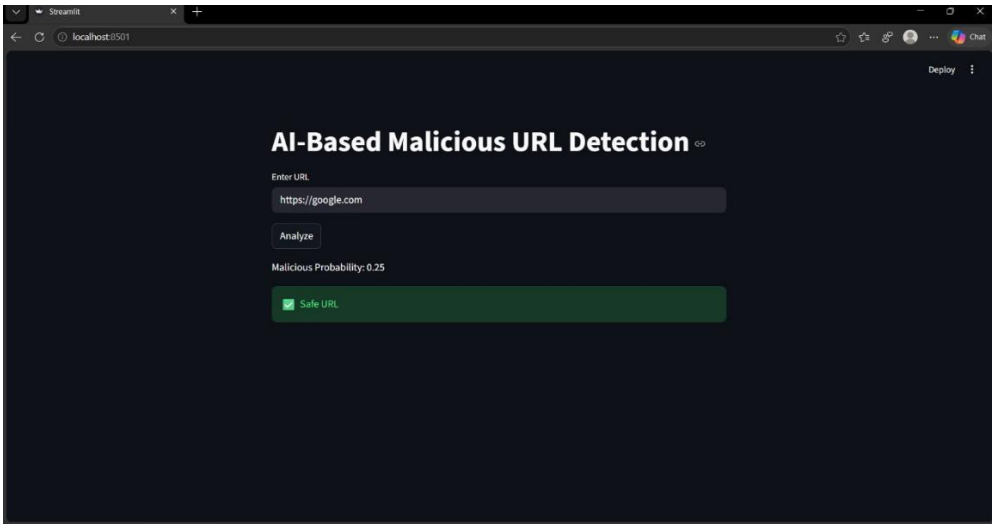


Figure 8.2 Check URL

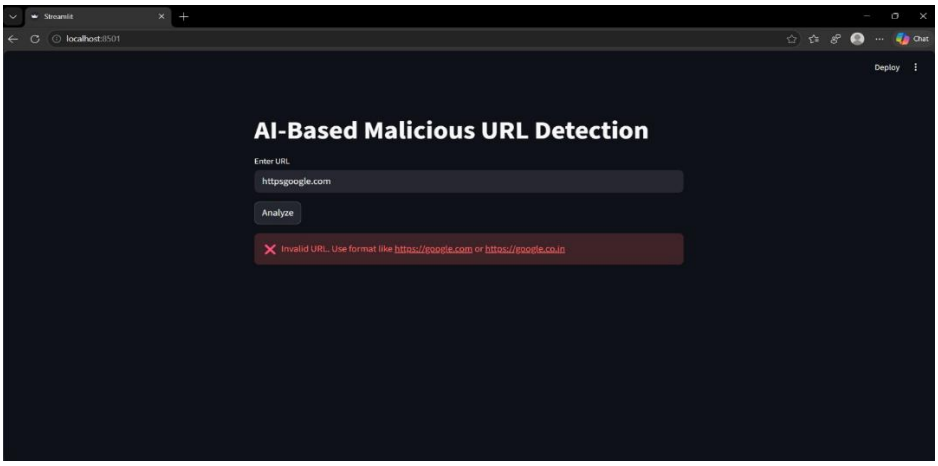


Figure 8.3 Check URL

CHAPTER 9
FUTURE SCOPE

9.1.FUTURE SCOPE

The future scope of the AI-Based Malicious URL Detection System includes several enhancements and advancements that can significantly improve its accuracy, scalability, and real-world applicability. With the continuous evolution of cyber threats and advancements in artificial intelligence, the system can be further developed in the following areas:

1. Enhanced Feature Engineering
Expanding feature sets beyond basic URL characteristics to include host-based, content-based, and behavioral features.
Incorporating domain age, SSL certificate details, DNS information, and webpage content analysis for improved detection accuracy.

2. Transformer-Based Deep Learning Models
Replacing or enhancing LSTM with advanced transformer models like BERT or DistilBERT.
Leveraging self-attention mechanisms to capture complex patterns and long-range dependencies in URLs.

3. Real-Time Threat Intelligence Integration
Integrating APIs such as VirusTotal, Google Safe Browsing, and OpenPhish.
Enhancing detection accuracy by combining AI predictions with real-time global threat data.

4. Browser Extension Deployment
Developing a browser extension for Chrome/Firefox to analyze URLs in real-time.
Providing instant warnings to users before accessing malicious websites.

5. Federated Learning for Privacy Preservation
Implementing federated learning to train models without sharing sensitive data.
Allowing organizations to collaboratively improve models while maintaining data privacy

6. Multi-Class Threat Classification

Extending the system from binary classification (safe/malicious) to multi-class classification.

Identifying specific threats such as phishing, malware, spam, or command-and-control attacks.

7. Cloud Deployment and Enterprise API

Deploying the system using cloud infrastructure and microservices architecture.

Providing REST APIs for integration with enterprise security systems like SIEM and firewalls.

8. Real-Time Scalable Detection System

Implementing scalable solutions using containerization and auto-scaling (e.g., Kubernetes).

Ensuring high performance and low latency for large-scale real-time URL analysis.

9. Advanced Explainable AI

Enhancing explainability using advanced SHAP visualizations and dashboards.

Helping security analysts understand model decisions more effectively.

10. Mobile Application Integration

Developing mobile applications for real-time URL scanning and alerts.

Providing user-friendly cybersecurity protection on smartphones.

CHAPTER 10

CONCLUSION

10.1.CONCLUSION

The proposed AI-Based Malicious URL Detection System represents a significant advancement in modern cybersecurity by effectively identifying and mitigating malicious web threats. It addresses the limitations of traditional detection systems by integrating multiple artificial intelligence techniques, including machine learning, deep learning, anomaly detection, and explainable AI, into a unified and robust framework.

One of the major strengths of the system lies in its ability to combine hybrid machine learning models with natural language processing techniques. The integration of Random Forest and Support Vector Machine through a soft voting mechanism significantly improves prediction accuracy and reliability. Additionally, the use of TF-IDF character-level feature extraction enhances the system's ability to detect subtle patterns within URLs that may indicate malicious intent.

The system further improves detection capabilities through the implementation of a deep learning model using LSTM, which captures sequential patterns in URLs. This enables the identification of complex and previously unseen attack patterns that may not be detected through traditional feature-based approaches. Moreover, the inclusion of an Isolation Forest-based anomaly detection module provides an effective mechanism for identifying zero-day attacks, thereby extending the system's coverage beyond known threats.

Another key contribution of the system is the integration of SHAP-based explainable AI techniques. This ensures transparency in model predictions by highlighting the contribution of individual features such as entropy, URL length, and suspicious keywords. Such explainability is crucial for building trust in AI-driven cybersecurity systems and assisting security analysts in decision-making.

From a performance perspective, the system achieves high accuracy, precision, recall, and F1-score, demonstrating its effectiveness in real-world scenarios. The hybrid model achieves superior performance compared to individual models, validating the effectiveness of ensemble learning. Additionally, the system is capable of real-time URL analysis with low latency, making it suitable for practical deployment.

The deployment of the system as a Streamlit web application further enhances its usability by providing an intuitive interface for real-time URL classification. This makes the system accessible to both technical and non-technical users, enabling broader adoption in various environments.

In conclusion, the AI-Based Malicious URL Detection System provides a comprehensive, scalable, and efficient solution for detecting malicious URLs. By combining multiple AI techniques, ensuring explainability, and enabling real-time deployment, the system not only addresses current cybersecurity challenges but also establishes a strong foundation for future advancements in intelligent threat detection.

CHAPTER11
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11.1.REFERENCES

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